Temperature prediction by gene expression programming

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Temperature prediction by gene expression programming

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***Abstract*—Air temperature is a crucial climatic component. Because of ever-changing weather, the prediction has evolved into a difficult feat. This research aims to predict the maximum temperature of the central region of Thailand by Gene expression programming (GEP). This technique is a fast and precise prediction technique results using climate measurements from previous years. The variables needed to construct the model are the daily maximum and minimum temperatures, relative humidity, and precipitation. Using Nash-Sutcliffe efficiency (NSE), Root mean square error (RMSE), and coefficient of determination (R^2) statistics, the performance of the GEP was examined. The results indicate that the GEP is reliable for predicting daily temperatures.**

Keywords— Climate Change, Weather, Artificial Intelligence, Modeling

I. INTRODUCTION

The monitoring and forecasting of temperature fluctuations are crucial to the study of future climate patterns. Global warming and climate change are currently global concerns due to their detrimental effects on human life [1]. It is essential for monitoring the climate, and droughts, forecasting severe weather, planning the energy, aviation industries, agriculture, and dispersing pollution [2]. Several techniques, including linear regression (LR), artificial neural networks (ANNs), Multi-Layer Perceptron Neural Networks (MLPNNs), auto-regression (AR), and Radial Basis Function networks (RBFNs), are used to predict atmospheric characteristics such as temperature, wind speed, precipitation, streamflow, rainfall-runoff, and meteorological pollution, among others [3-7]. In weather forecasting, the forecasting of temperature variation conditions is crucial for a variety of applications. Temperature prediction is the essential service supplied by meteorologists to protect the lives and property of citizens in each country [2]. Using ANNs, numerous forecasting algorithms have been created in recent years. Traditional techniques can tackle complex, nonlinear issues more effectively than neural network techniques. However, systems constructed with the neural network model include disadvantages such as local minimums and model overfitting [8]. GEP was first proposed by Ferreira (2001) based on Darwin's theory of natural selection[9]. This study presents a concept for developing a prediction model for temperature utilizing the GEP and modifying the original algorithm to increase convergence time and prediction model accuracy. This study is structured as follows: Part 2, the concept for Temperature prediction utilizing the GEP technique and its application for estimating the temperature in

Bangkok, Thailand, will be provided. Also, it includes the study area and dataset used. In part 3, the results, recommendations, and future work will be provided.

1. Methodology

A. Study Area

Thailand's climate is tropical. It is always hot, especially from March through May, with April being the hottest month (30°C/86°F). It is in Thailand's central region, on the vast, flat map of the Chao Phraya River, at latitude 13.45° and longitude 100.35° E. There are three seasons in a year: rainy (April to October), cold weather (November to January), or hot (June to August) (February–March)[10].

B. Datasets

From 2018 to 2022, historical metrological data from 19 stations were gathered. The span from 1 January 2018 until 31 December 2020 was chosen to train the GEP model, whereas from 1 January 2021 to 10 September 2022 served as the testing set. This study focuses solely on the central region of Thailand. Hence a total of 19 stations within the central region area were chosen, while all other stations situated outside the central region were disregarded, with each chosen station representing approximately 91,799 km^2 of land. The collected meteorological data included daily relative humidity measurements, maximum temperature, minimum temperature, and rainfall data.

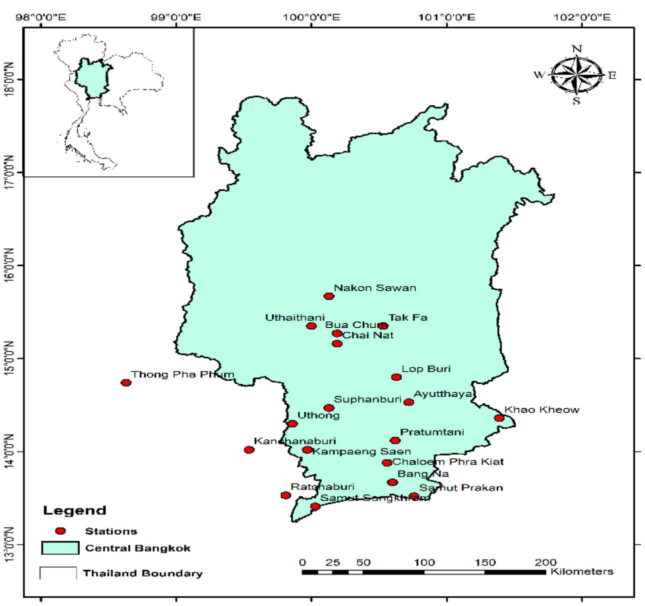

Fig 1 Central region of Thailand

C. GEP

It was first proposed by Ferreira (2001) and built on the Darwinian Evolution Theory [9]. The GEP combines genetic algorithm (GA) and genetic programming (GP) techniques. In the GEP, entities such as genomes are encoded as linear sequences of fixed length [11, 12]. GP is a technique that permits problems to be solved by self-generating expressions and algorithms. These algorithms are encoded as a tree-like structure with their terminals and functions serving as leaves (nodes). GP primarily programmed the GAs using tree structures. For general practitioner-based problem resolution, there are five initial phases.

The function of (1) terminal set and (2) function set (3) evaluation of fitness (4) management of the race using numerical parameters and quantitative variables (5) The prerequisite of describing a result and concluding a results sequence [13]. Individually classifying the collection of terminals used in the computer programs is the initial key stage in preparing the GP hypothesis. Independent variables, state variables, and functions without arguments constitute the primary types of terminal sets. The second crucial phase describes the functions set, testing functions (IF and CASE expressions) and Boolean functions (AND, OR NOT). The third primary phase is fitness measurement, which assesses how effectively a program addresses a specific issue. Terminals and functions combine to create the intersections of the tree. The choice of specific variables to control and regulate the run constitutes the fourth primary phase. Population size and crossover rate are the factors of control. The fifth stage of the run is completed. Generally, a model is deemed satisfactory if its outcomes and the data are identical. After specifying terminal and non-terminal operators, the types can be configured [11, 12]. Darwin's theory of evolution governs the formation of automatic programs; after subsequent generations, old trees build new trees via copy, crossover, and mutation [14]. Based on natural selection, superior trees are more likely to be selected for the next generation. Consequently, a well-modified tree is obtained when a stochastic process is implemented [15]. The flow diagram of the working of GEP to predict daily maximum temperature is shown in Fig 2.

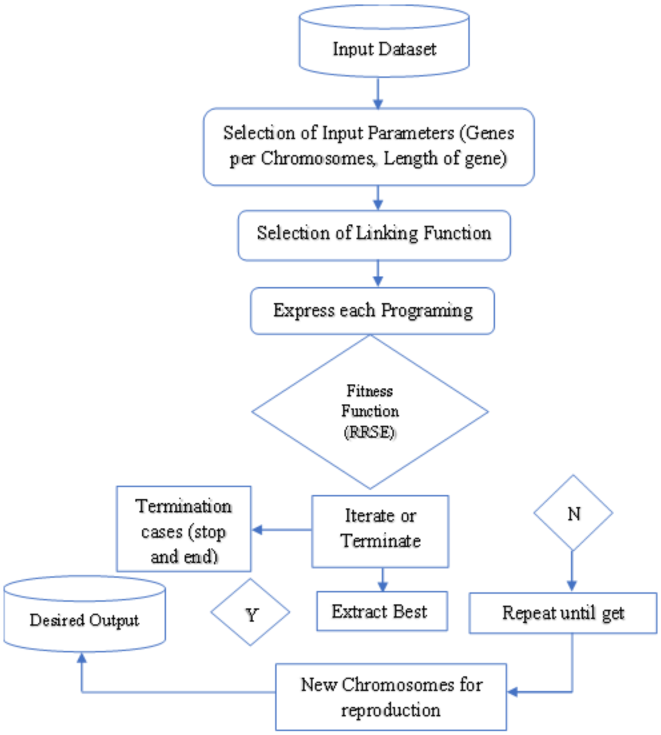

Fig 2 Working layout of Gene Expression Programming

D. Performance Assessment

In this study, three performance assessment parameters were used to evaluate the efficiency of the model by the R^2, NSE, and RMSE, which are defined as:

$$R^2 = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{n[\Sigma x2 - (\Sigma x2)][\Sigma y2 - (\Sigma y2)]}} \quad (1)$$

$$\text{NSE} = 1 - \frac{\sum_{i=1}^{N}(Tobs - Tpre)^2}{\sum_{i=1}^{N}(Tobs - Tmean)^2} \quad (2)$$

$$\text{RMSE} = \frac{\sqrt{\sum_{i=1}^{N}(Tobs - Tpre)^2}}{N} \quad (3)$$

Tobs and Tpre are the observed and predicted daily maximum temperatures, respectively, while Tmean is the mean of observed daily maximum temperature.

III. Results and Discussion

We examined our suggested GEP model using the previously mentioned 5-year data. This dataset contains January maximum and minimum temperatures, relative humidity, and precipitation for CL (2018 to 2022). The selected dataset was divided into two groups, the training group, which included 70% of the data, and the test group, which contained 30% of the data, to evaluate the generalization capability of the stimulation model after the training phase. The training dataset is utilized to build the model, the validation set to optimize the model's parameters, and the testing dataset to evaluate the model. We utilized R^2, NSE, and RMSE as measures of GEP error. The results are shown in Table 1. The R^2 of GEP was 0.62 during the training period, which improved in the testing period to 0.74. Overall results of RMSE also improved in the testing period from 1.68 to 0.85.

Table 1 Performance Evaluation of GEP during training and testing period

Period	**R^2**	**RMSE**	**NSE**
Training	0.62	1.68	1
Testing	0.74	0.85	1

Fig 3 (a) and (b) show the standard Deviation (SD) Mean and coefficient of variation (CV) results obtained by the GEP and observed maximum temperature for the selected region during training and testing periods, respectively. The actual value Mean of the maximum temperature for the training period is 31.51068, and the Predicted value Mean of the maximum temperature is 31.56531. Similarly, 32.74977 and 32.77410882 for the testing, respectively.

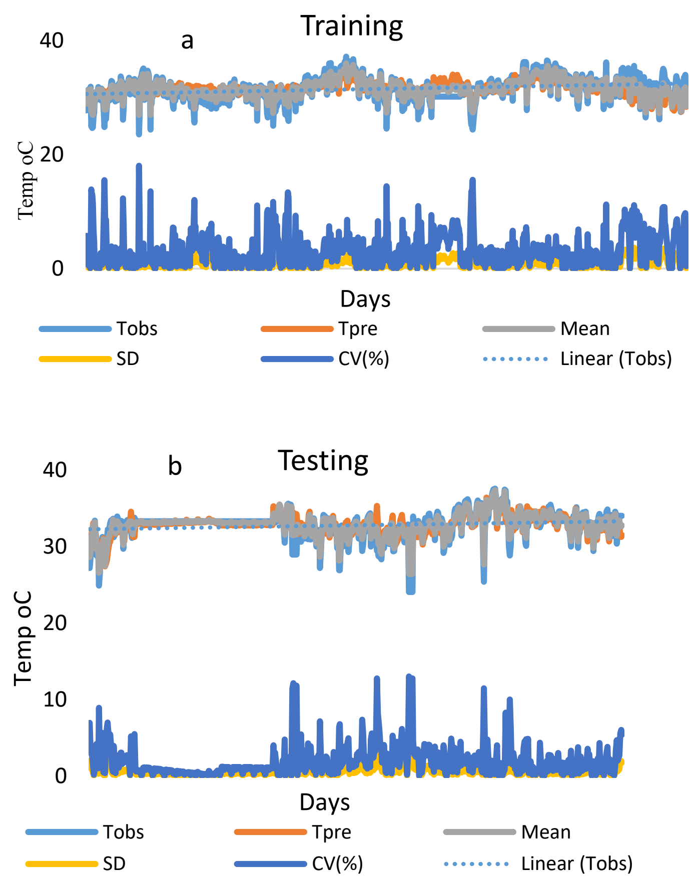

Fig 3 (a) and (b) Statistics of GEP during training and testing periods.

The observed and predicted daily maximum temperature during the training and testing period is shown in Fig 4 (a) and (b).

The implementation of the GEP model was much improved during the testing period. Hence, overall observations indicate that this GEP technique worked well in predicting Daily Maximum Temperature. Fig 5 (a) and (b) Daily Maximum temperature training and testing residual data is shown. The residual variance after modeling during the training of GEP was 3.3566761, which was 4.7254587 before scoring data. On the other hand, 3.5344772 variances were found before scoring data during testing, and the residual variance after modeling was 1.7053332. The proportion of variance explained by the model (R^2) was 0.51751 (51.751%). The outcomes of these parameters showed that the prediction capability of GEP is reliable for the central region of Thailand.

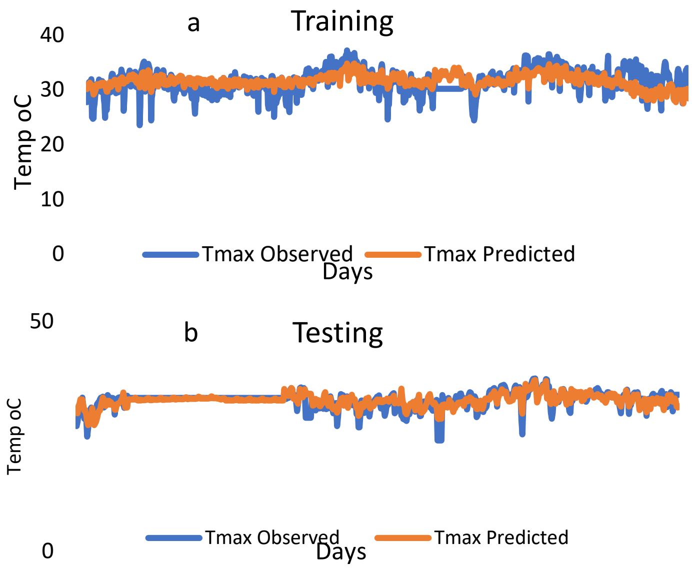

Fig 4 (a) and (b) Observed and Predicted Temperatures

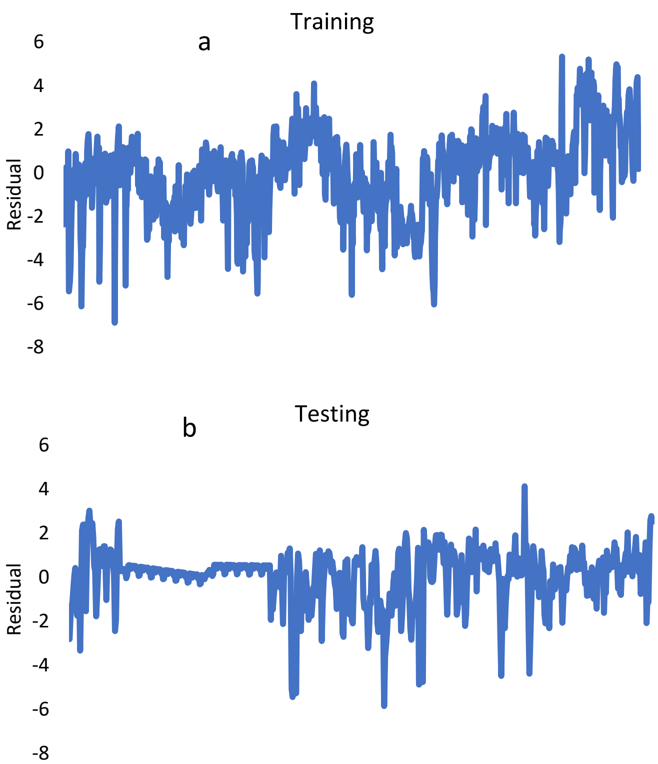

Fig 5 (a) and (b) Daily Maximum temperature training and testing residual data

IV. CONCLUSION

This research presents the use of the GEP method for temperature prediction. The performance of GEP was evaluated using various statistical metrics. The results indicate that SVM performs better in Thailand's tropical region. It was also noted that the choice of input parameters substantially impacts the model's performance. The results of these parameters demonstrated that the GEP's capacity to predict outcomes is accurate for Thailand's central area. GEP can be used to replace some neural network-based models in weather prediction applications if the correct input datasets are used.

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